

The Tokyo raid

The Pearl Harbor raid and the setbacks that followed in its wake left Americans thirsting for vengeance and in need of a lift to morale. Looking for a spectacular way of hitting back at the Japanese, the Americans hatched the plan of a carrier-launched bombing raid on Tokyo. Because no carrier aircraft had the range to strike Japan from far enough out, the USAAF was called on to provide North American B-25 Mitchell bombers. No one had ever thought that one of these aircraft could be flown off a carrier flight deck, but now they would have to.

Volunteer USAAF crews led by the renowned Lieutenant-Colonel James Doolittle were put through an intensive course of training in short takeoffs and low-level, long-distance flight before sailing from San Francisco on the carrier *Hornet*

“The raid had a great many things going for it, but... the biggest thing was morale for the American people.”

REAR ADMIRAL HENRY L. MILLER
ON THE DOOLITTLE RAID

in April 1942. Sixteen bombers were tethered on the flight deck because they were too big to be stowed below, leaving no room for the *Hornet*'s own aircraft to take off or land. The carrier *Enterprise* joined up to provide air cover. The plan was for the bombers to take off late on April 18, when the *Hornet* would be about 400 miles (640km) from Tokyo. They would raid the city under cover of darkness and fly on to land at Chuchow airfield in China, which was held by friendly forces. But early on the morning of the 18th, the task force was spotted by Japanese patrol boats. It was decided to launch the bombers immediately, although Tokyo was now 650 miles (1,050km) away and the raiders would arrive in daylight.

Doolittle was the first to take off. With the aircraft laden with bombs and extra fuel, and the carrier pitching in a heavy sea, conditions were hardly ideal for the first sea launching of a B-25. Fortunately there was a 30-knot wind, which, with the 20-knot progress of the carrier, gave a

TAKING OFF FOR TOKYO

Doolittle takes off from USS Hornet on April 18, 1942. Each B-25 had to lift off just as the pitching deck swung up toward the crest of a wave.

THE THACH WEAVE

EVEN THE BEST US Navy pilots were disturbed by having to fly against the Mitsubishi Zero fighter, which could outperform the Grumman F4F Wildcat. Fortunately, the navy had a number of exceptional pilots who developed tactics to counter the Zero. Jimmy Thach, the commander of Fighting Squadron 3, had the idea of abandoning the three-aircraft V formation – a leader and two wingmen – that was standard for US Navy fighters and previously had been for the RAF. Instead, he had his pilots adopt a four-plane combat unit made up of two two-plane sections, the same solution that the Luftwaffe Messerschmitt pilots had come up with.

Using the four-aircraft formation, Thach developed a maneuver specifically designed to negate the Zero's superiority in performance, which would theoretically give it the upper hand in any combat situation. Known as the “Thach Weave,” the maneuver went like this. The two pairs of Wildcats flew with just enough air between them for a tight turn. They watched each other's tails. If a Zero came down behind one of the left-hand pair of aircraft, for example, the right-hand pair would wait until the Japanese aircraft was almost in shooting range and then turn sharply toward their

FLYING FOURS

US Navy Grumman F4F Wildcats fly in the four-plane fighting formation pioneered by Jimmy Thach. Each combat unit of four aircraft is made up of two two-plane sections. A section consists of a leader and his wingman.

colleagues. Seeing the turn, the targeted Wildcat would know he had a Zero on his tail and dive down and to his right. The right-hand pair of Wildcats would get a side shot at the Zero if he pulled out or a head-on shot if he turned to keep his target in his sights. Thach had the chance to implement this tactic at the Battle of Midway.

wind speed of 50 knots to aid liftoff. Doolittle made it look easy and the other 15 somehow got off behind him with only one minor incident.

The sea-skimming flight to Tokyo took four hours and achieved total surprise. The B-25s did not have a sufficient bombload to cause much damage, but their sudden appearance was a severe shock to the Japanese. The aftermath to the raid unfortunately did not go as planned.

Most of the bombers ran out of fuel and all were lost, as crews either baled out or crash-landed. Of the 80 aircrew, 73 survived, including Doolittle who returned to make a further distinguished contribution to World War II as a commander in the European theater. Of little tactical significance, the Doolittle raid shocked the Japanese into rushing their Pacific-expansion plans.

The Battle of Midway

The instructions to the US Pacific Fleet in 1942 were to “hold what you've got and hit them when you can.” There were only six full-size American carriers when the war started and their numbers were soon reduced by enemy action. The Japanese enjoyed superiority both in the quantity and overall quality of their naval aircraft and pilots. The Battle of the Coral Sea was a setback for Japan – a lesson that the Americans were still in the ring and fighting. But Japanese naval commanders

